Module 3:

Environmental Pollution

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A Place to Learn, A Chance to Grow



EVS- An Insight

Key Environmental Problems and Sustainability

Global environmental issues, socio-environmental interaction; impact of consumerism and global economy; Sustainability: Fundamentals, science behind sustainability, challenges, potential; role of technology, policies.

Ecosystem and Biodiversity

Ecosystem: functions and services, Nutrient, phosphorous, carbon, nitrogen, cycles; Effect of human activities on these cycles. Threats to biodiversity: conservation strategies, sustainable solutions.

Environmental pollution

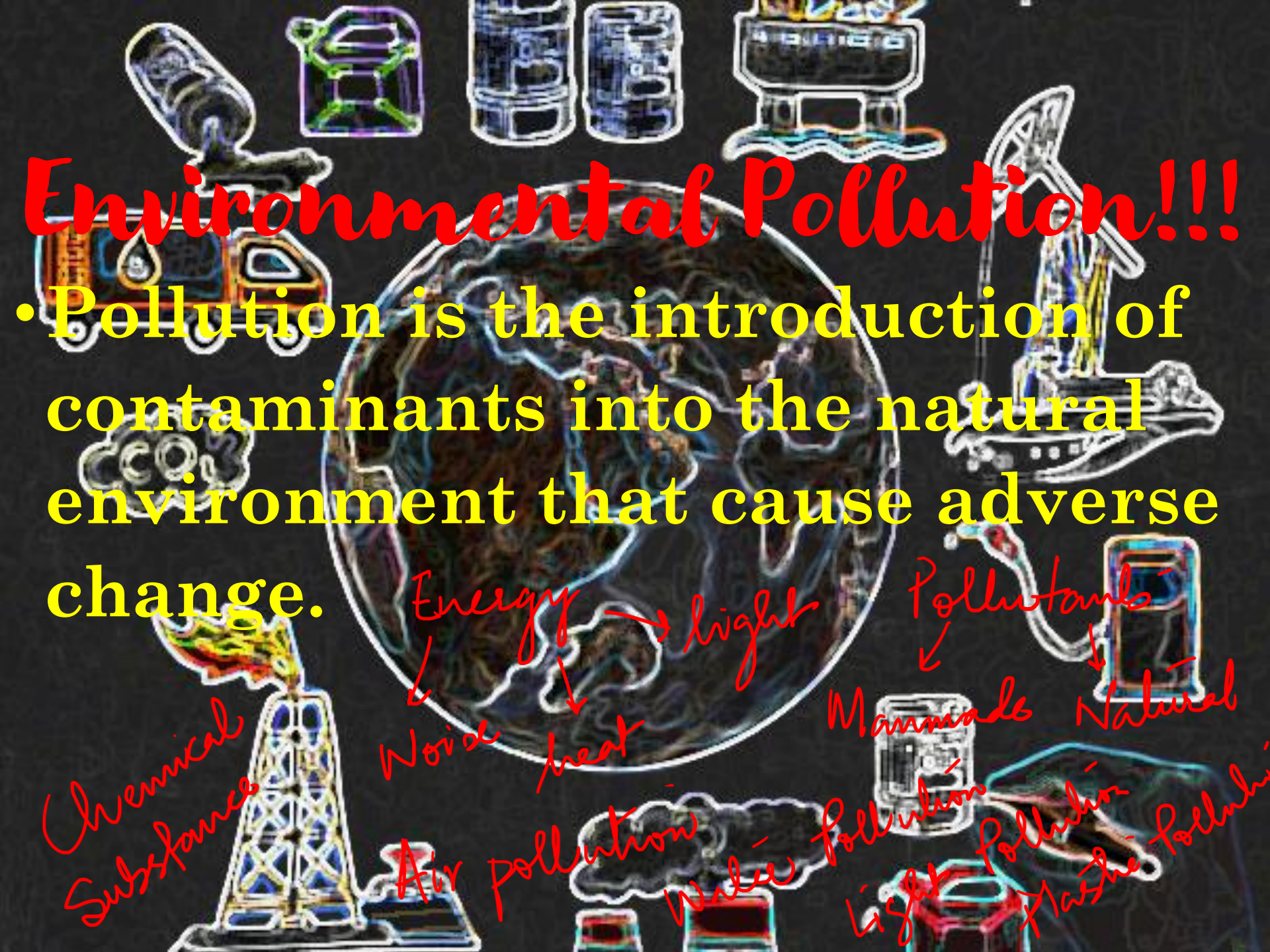
Pollution sources, effects and mitigation, conservation. Water footprint and virtual water. Sustainable solutions. Environmental Protection Act – Air, water, forest and wild life. Interactional protocols and agreements. Resource recycling and sustainability.

Human Population and Environment

Impact of population age structure, human population growth, Sustaining human societies: Economics, environment, policies and education.

Energy Sources and Sustainability

Conventional sources, non-conventional sources, natural resources and economics, clean coal technology, sequestration methods, energy conservation and sustainability, policies, acts, examples.

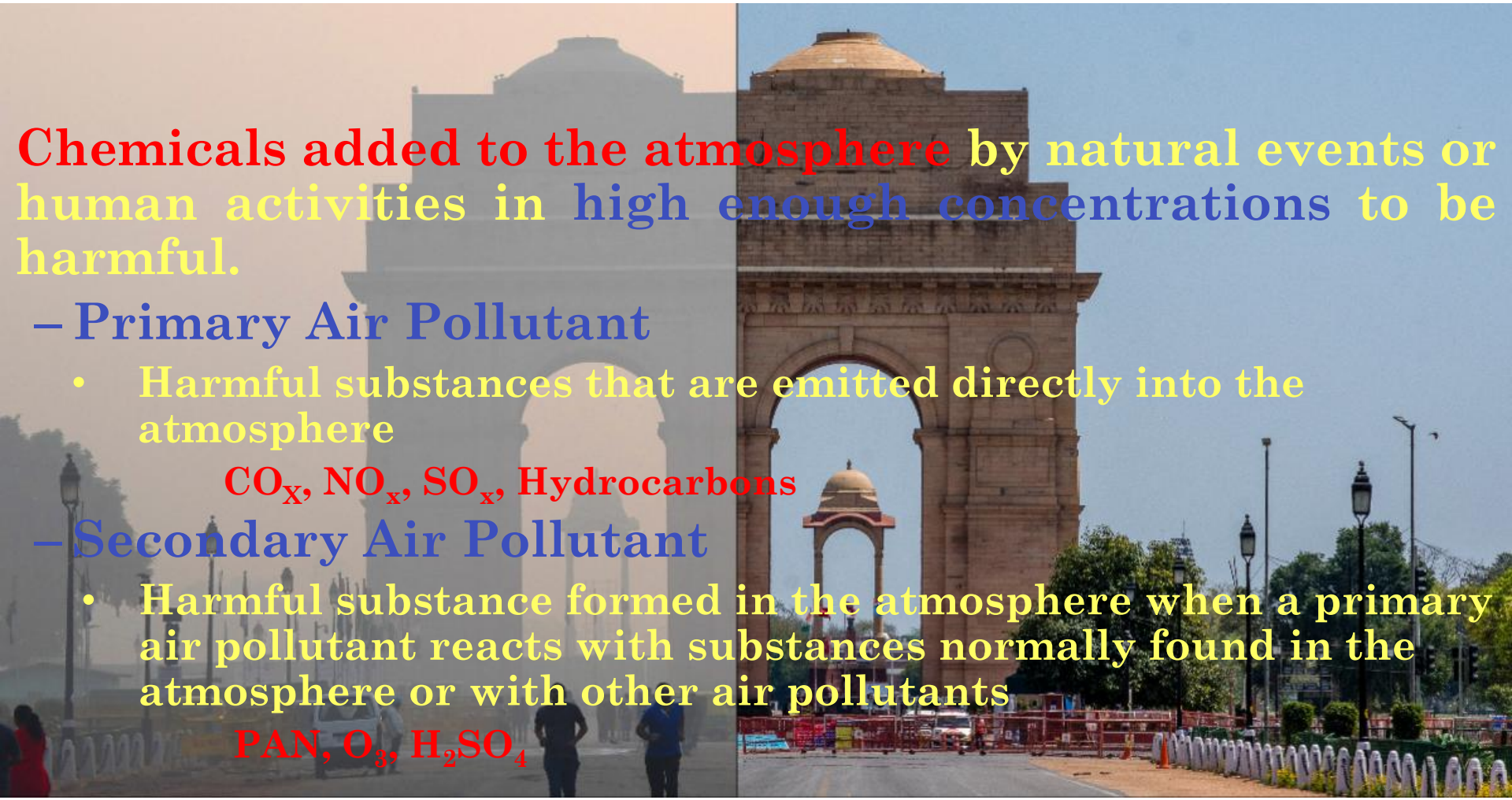


- Pollution is the introduction of contaminants into the natural environment that cause adverse change.

Air Pollution 101



Air Pollution



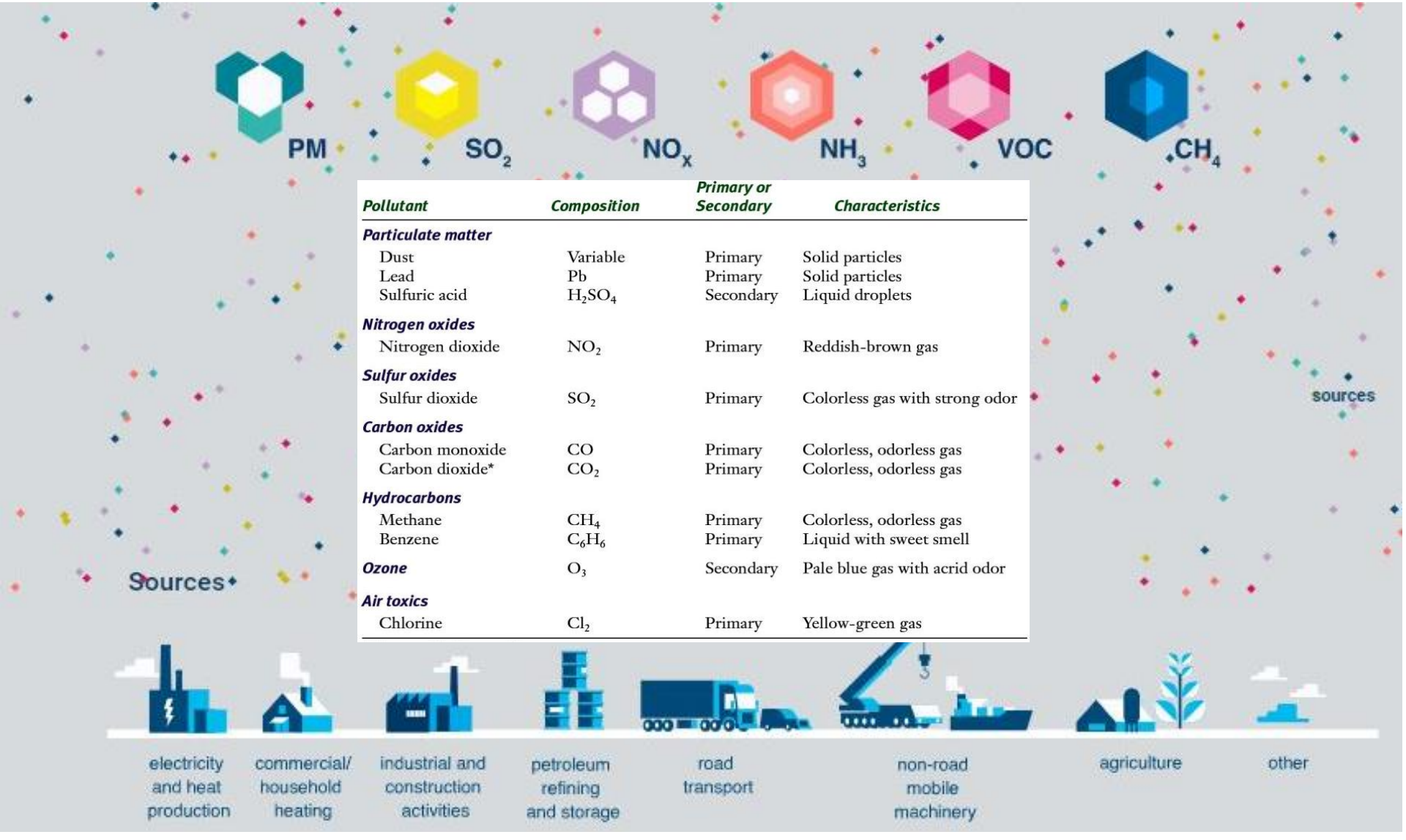
Chemicals added to the atmosphere by natural events or human activities in high enough concentrations to be harmful.

- **Primary Air Pollutant**
  - **Harmful substances that are emitted directly into the atmosphere**

    CO_X, NO_x, SO_x, Hydrocarbons
- **Secondary Air Pollutant**
  - **Harmful substance formed in the atmosphere when a primary air pollutant reacts with substances normally found in the atmosphere or with other air pollutants**

    PAN, O_3, H_2SO_4

Major Air Pollutants



Pollutant	Composition	Primary or Secondary	Characteristics
Particulate matter			
Dust	Variable	Primary	Solid particles
Lead	Pb	Primary	Solid particles
Sulfuric acid	H_2SO_4	Secondary	Liquid droplets
Nitrogen oxides			
Nitrogen dioxide	NO_2	Primary	Reddish-brown gas
Sulfur oxides			
Sulfur dioxide	SO_2	Primary	Colorless gas with strong odor
Carbon oxides			
Carbon monoxide	CO	Primary	Colorless, odorless gas
Carbon dioxide*	CO_2	Primary	Colorless, odorless gas
Hydrocarbons			
Methane	CH_4	Primary	Colorless, odorless gas
Benzene	C_6H_6	Primary	Liquid with sweet smell
Ozone	O_3	Secondary	Pale blue gas with acrid odor
Air toxics			
Chlorine	Cl_2	Primary	Yellow-green gas

Sources

Natural Sources :

Forest Fire

Sea Salt sprays

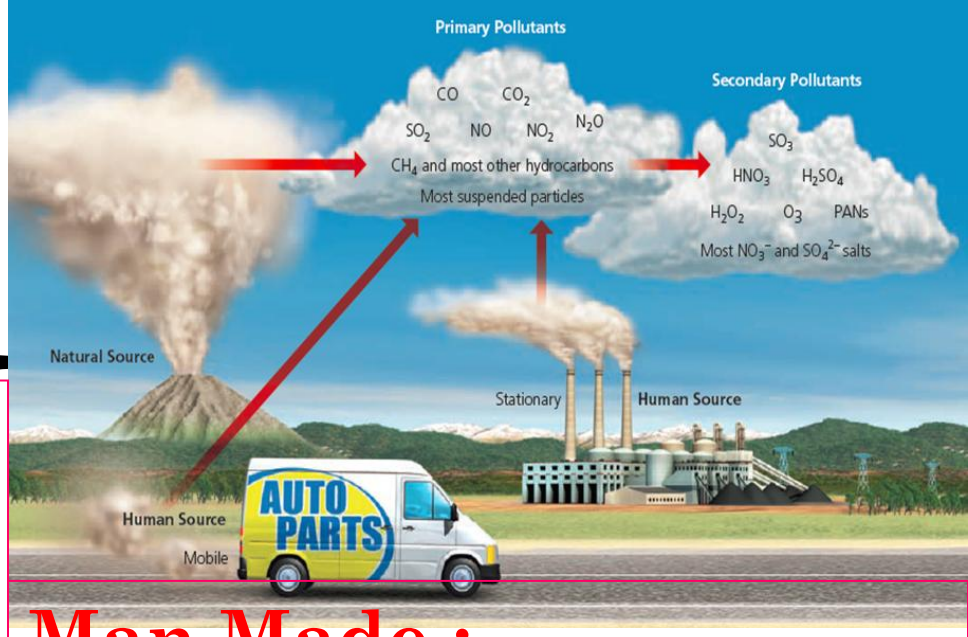
Methane, emitted by the digestion of food by animals, for example cattle

Smoke and carbon monoxide from wildfires

Volcanic activity, which produces sulfur, chlorine, and ash particulates

pollen grains of flowers, spores

Radon gas from radioactive decay within the Earth's crust



Man Made :

Vehicular emission,

thermal power plant – fly ash and SO_2

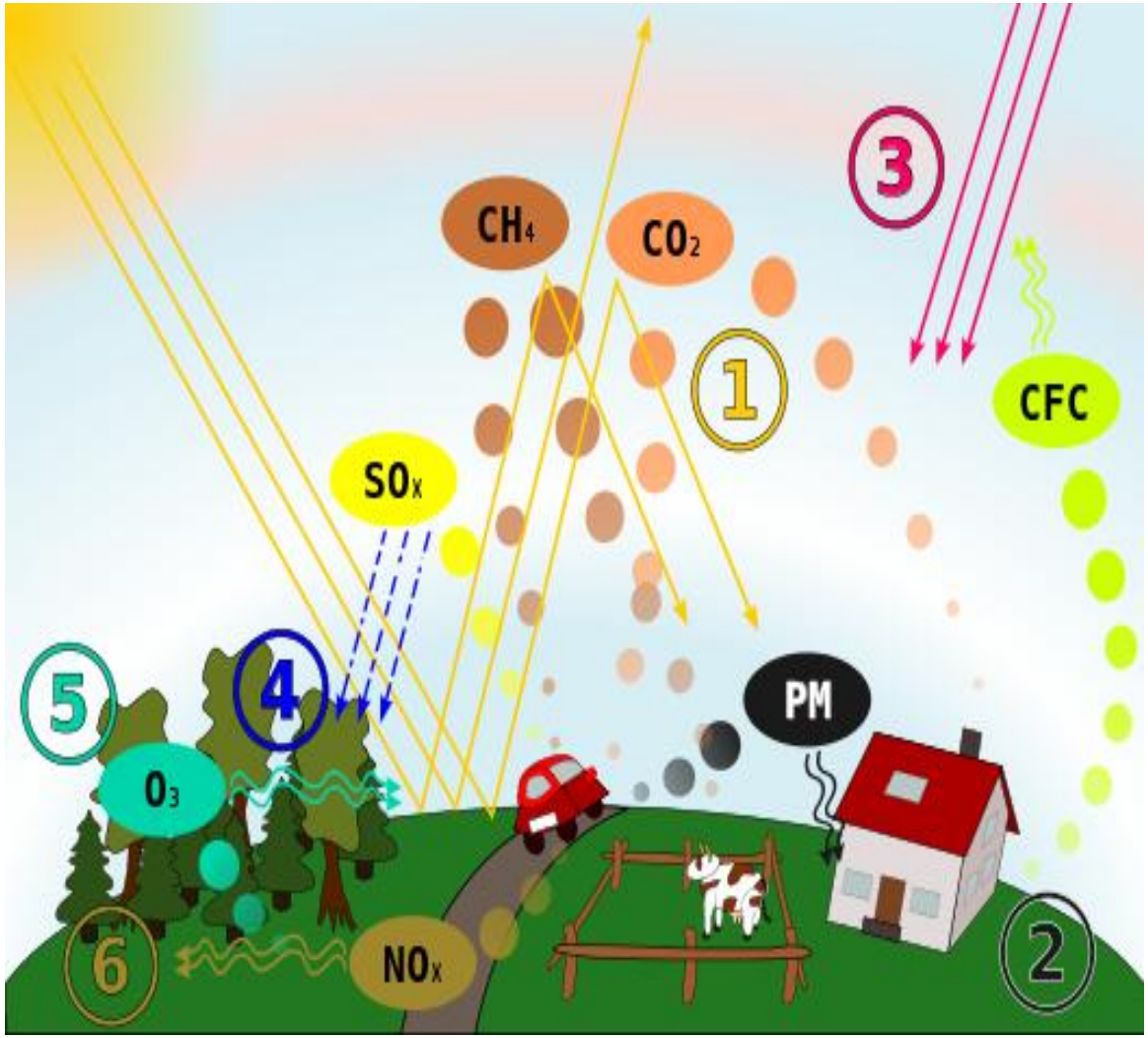
Fossil fuel burning

refineries

chemical industries

Note: automobiles releases ~77% CO, ~8% NO_x, ~14% HC, Heavy duty diesel vehicles spew more NO_x and SPM than petrol vehicles.

Schematic drawing, causes and effects of air pollution



1. Greenhouse effect
2. Particulate contamination
3. Increased UV radiation
4. Acid rain
5. Increased ground-level ozone concentration
6. Increased levels of nitrogen oxides.

Major Classes of Air Pollutants (Self-study)

- Particulate Matter
- Nitrogen Oxides
- Sulfur Oxides
- Carbon Oxides
- Hydrocarbons
- Ozone

AQI, Movement of Pollutants & Factors that Effect Air Quality

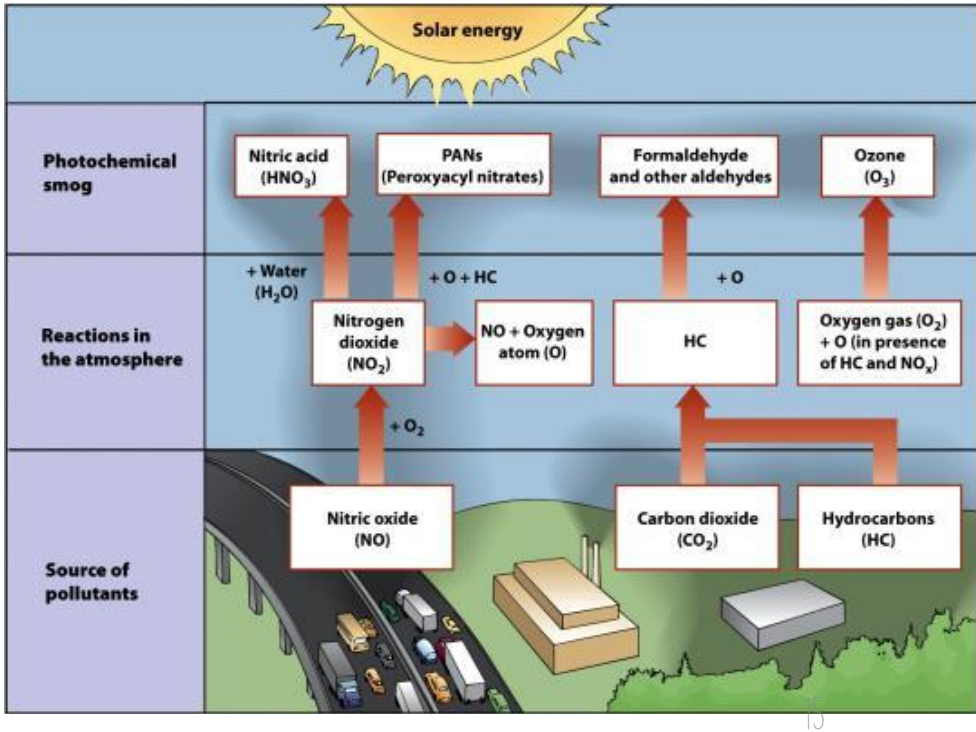
(a)Factors that Affect Air Quality:
http://has.concord.org/air-pollution.html

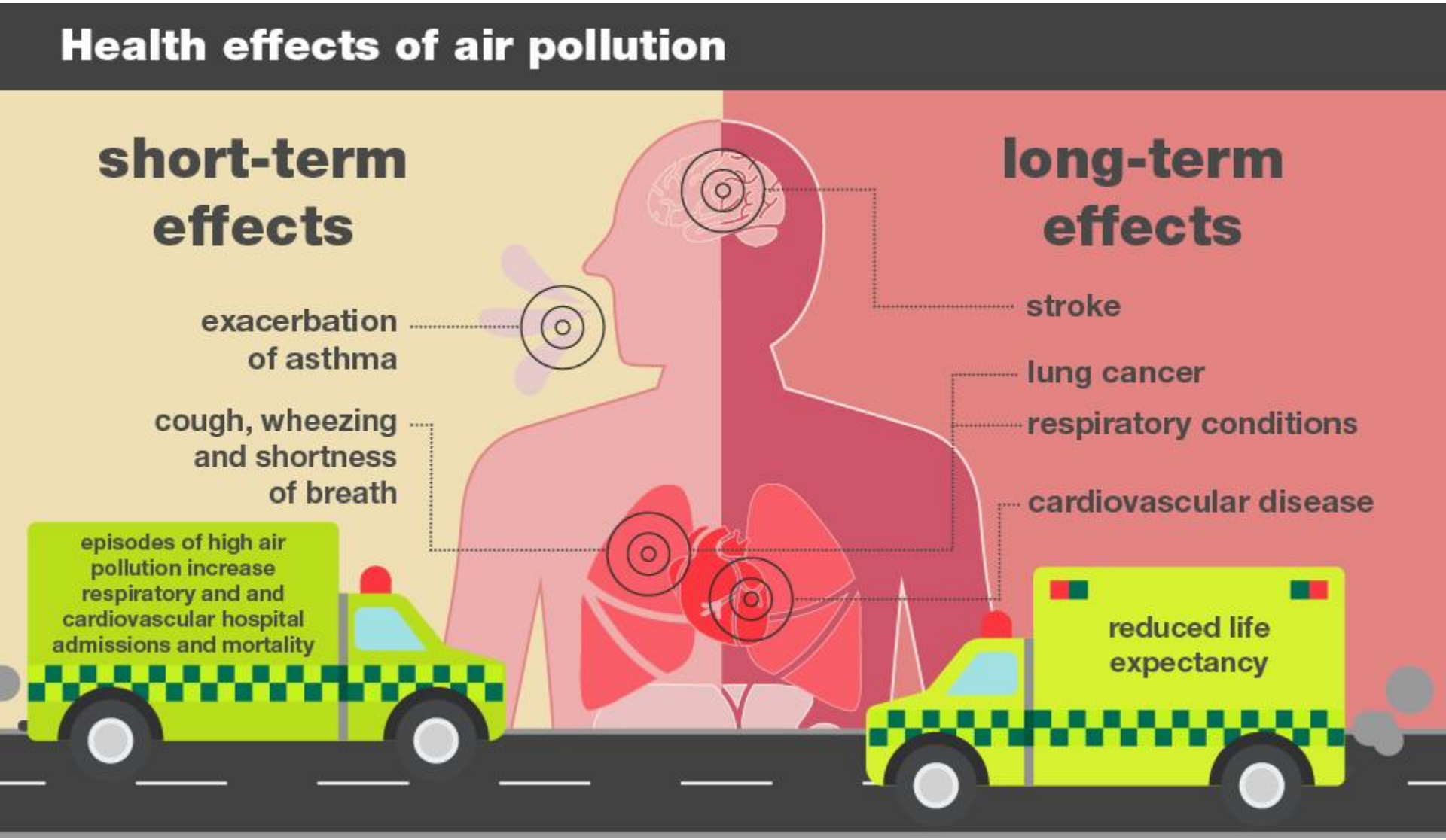
(b) Movement of Pollutants:
http://has.concord.org/air-pollution-aerial.html

(c) Air Quality Index:
https://waqi.info/

Effect on environment: Photochemical Smog "Summer Smog"

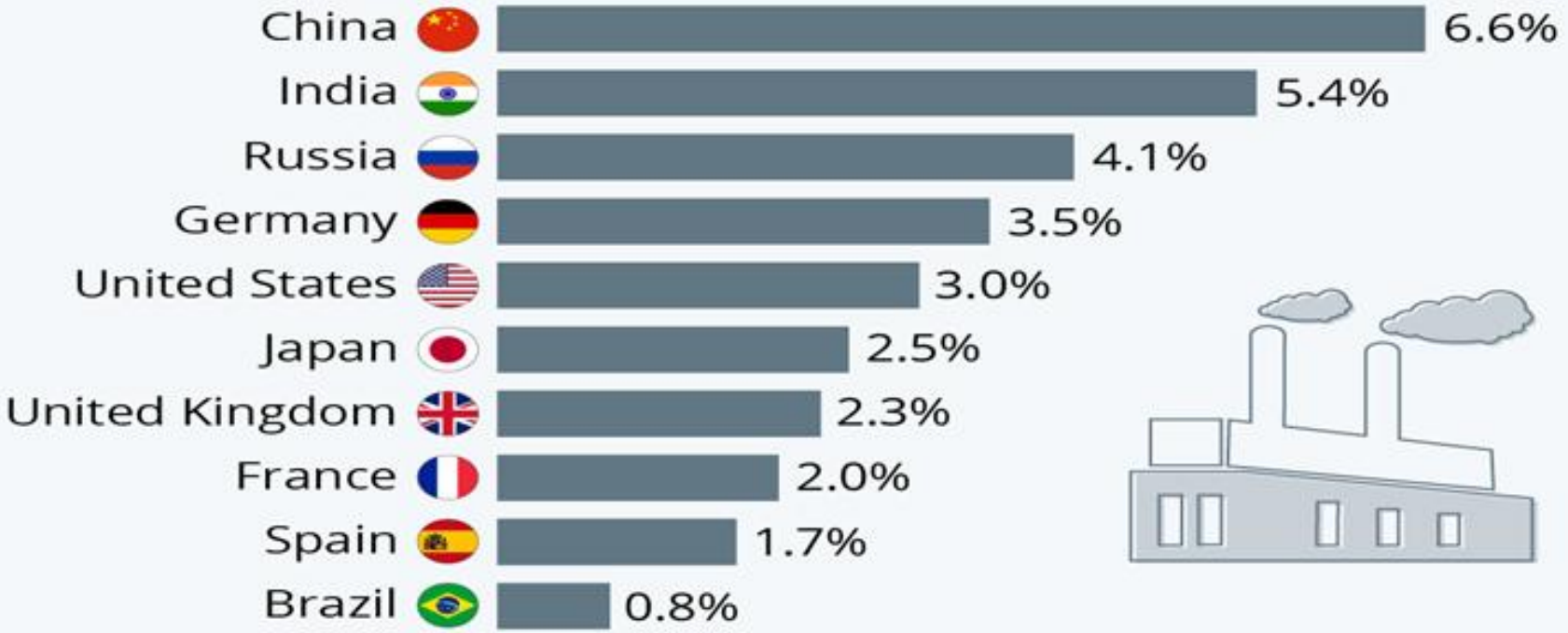
- **Brownish-orange haze formed by chemical reactions involving sunlight, nitrogen oxide, and hydrocarbons**
- **Sunlight Plus smoke from Cars/Industries Equals Photochemical Smog**
- A *photochemical reaction* is any chemical reaction activated by light.
- **Photochemical smog** is a mixture of primary and secondary pollutants formed under the influence of UV radiation from the sun.




Health effects of air pollution
short-term effects
exacerbation of asthma
cough, wheezing and shortness of breath
episodes of high air pollution increase respiratory and and cardiovascular hospital admissions and mortality
long-term effects
stroke
lung cancer
respiratory conditions
cardiovascular disease
reduced life expectancy

The Economic Burden Of Air Pollution

Economic costs of air pollution from fossil fuels as a share of GDP in 2018



Sources: Greenpeace, Center for Research on Energy and Clean Air

statista

Effect on Plants

- The pollutants present in the air diffuses through the leaf pores (stomata) and destroy the chlorophyll and affect the photosynthesis.
- During the daytime, the pores are wide open to facilitate photosynthesis and hence pollutants easily diffuse and affects the plants.
- They also erode the cuticle (waxy layer on the leaves) which prevents the excess water loss.
- It causes the following diseases

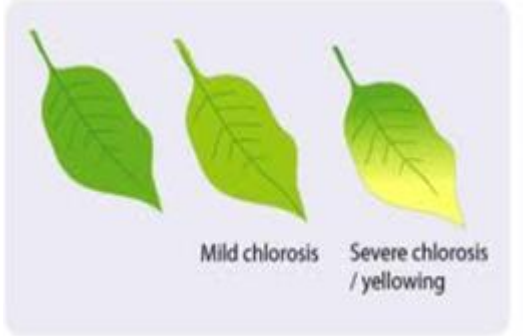
  (i) necrosis – dead areas of leaf

  (ii) chlorosis – yellowing of leaf due to loss of Chlorophyll

  (iii) epinasty – downward curling of leaf

  (iv) abscission – dropping of leaves







Effects on aquatic life



- High acidity in fresh water lakes.
- The most serious chronic effect of increased acidity in surface waters appears to be interference with the fish's reproductive cycle. Calcium levels in the female fish may be lowered to the point where she cannot produce eggs or the eggs fail to pass from the ovaries or if fertilized, the eggs and/or larvae develop abnormally.
- Some freshwater lakes have experienced total fish death.
- Due to **acidification**, 18 **Norwegian** stocks of **Atlantic salmon** are extinct and an additional 8 are threatened.



Effects on materials

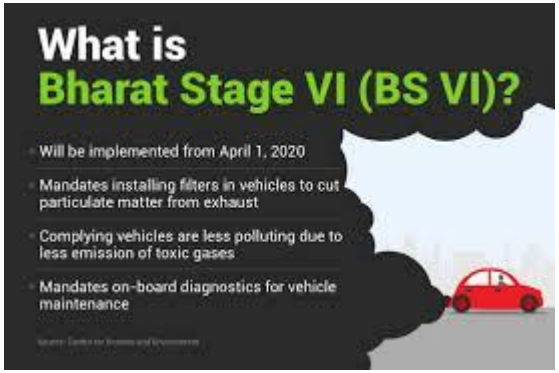
- SO_2: along with moisture can promote corrosion of metallic surfaces due to formation of sulfuric acid
- Sulfuric acid: can damage buildings and cause disfigurement of statues made of marble and limestone, damages leather binding of books, makes book pages brittle
- Ozone in the atmosphere can cause cracking of rubber
- Nylon stockings are weakened and damaged
- Cotton and rayon fibres damaged by oxides of nitrogen and ozone

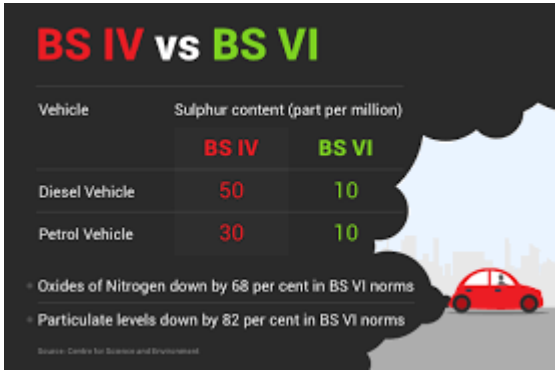


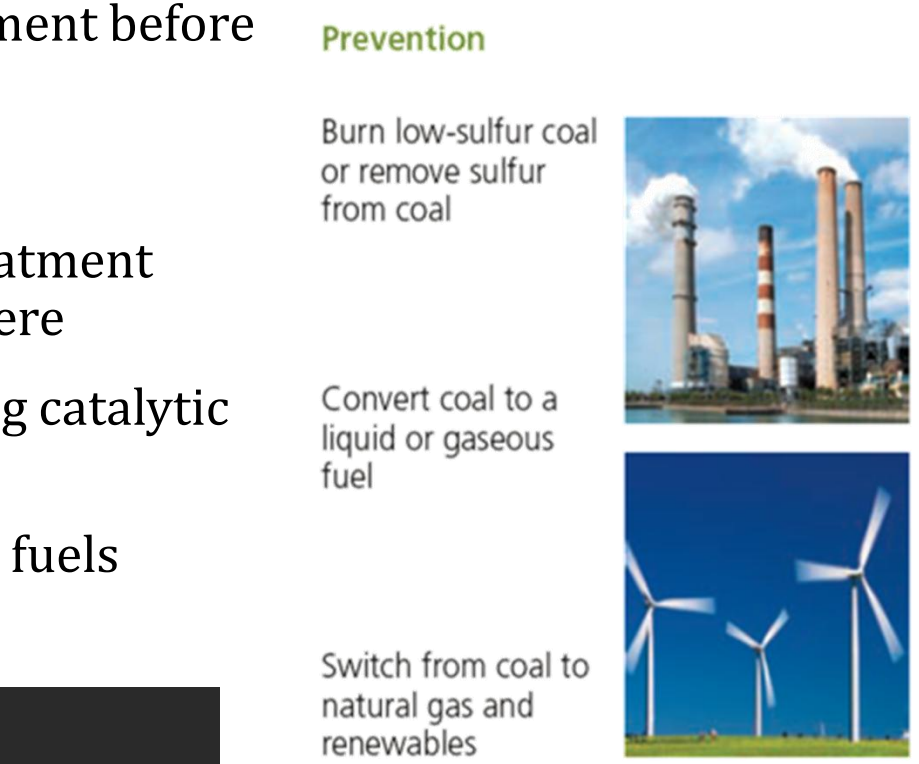


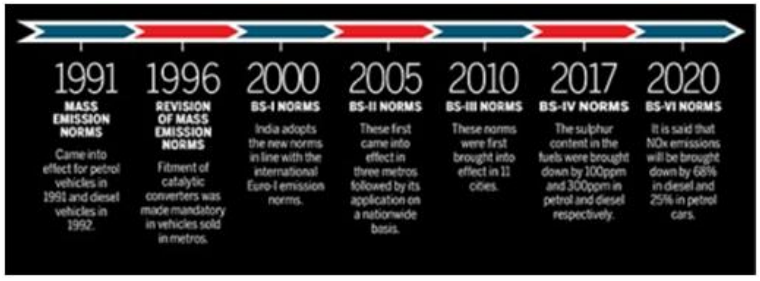
Prevention/mitigation

- Proper environmental impact assessment before setting up an industry
- Using low sulfur content coal
- Using exhaust gas analysis and its treatment before being let out into the atmosphere
- Regular tuning of engine and installing catalytic converters for the oxidation of CO
- Using mass transport and using clean fuels
- Planting more trees









PROTECT YOUR FAMILY FROM INDOOR AIR POLLUTANTS

WE KNOW THE SECRET TO HEALTHY LIVING

Dangerous Gasses

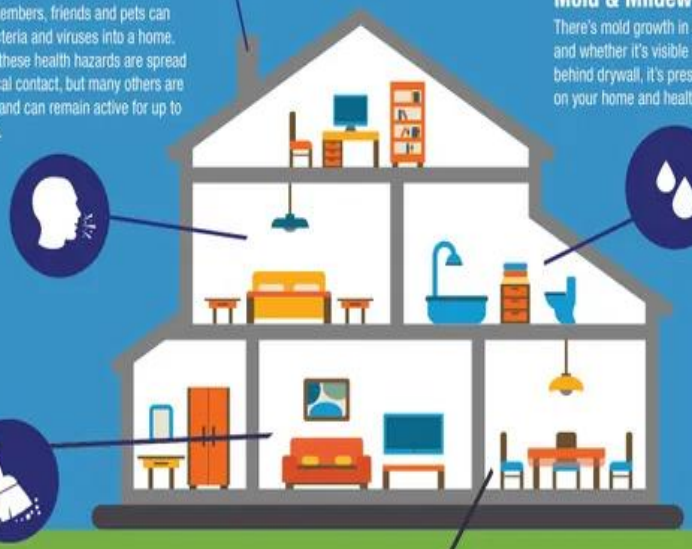
Carbon monoxide (CO) and radon are hazardous gases made even more dangerous because they're invisible and odorless.

Viruses & Bacteria

Family members, friends and pets can bring bacteria and viruses into a home. Some of these health hazards are spread by physical contact, but many others are airborne and can remain active for up to 24 hours.

Mold & Mildew

There's mold growth in over 50% of households, and whether it's visible on surfaces or hidden behind drywall, it's presence can wreak havoc on your home and health.



Allergens & Particulates

In a single year, a 6-room house collects an average of 40 pounds of dust, pet dander, pollen & other harmful allergens that can act as a trigger for people who suffer from allergies or asthma.

Odors & Toxic Compounds

Also referred to as VOCs, volatile organic compounds are emitted into the air as gases and come from common household products such as cleaners, paints, aerosol sprays, disinfectants, air fresheners, as well as building materials and furniture.

BREATHE EASIER:
Improving Indoor Air Quality in Your Living Room

POOR INDOOR AIR QUALITY CAN WORSEN ASTHMA AND ALLERGIES

Choose leather or vinyl furniture.

Hang blinds or washable curtains.

Don't smoke in the house or use the fireplace.

DON'T USE throw pillows.

CLEAN UP clutter.

Use washable area rugs. **WASH** them weekly.

Remove scented candles and potpourri.

Replace carpets with solid surface flooring with low VOCs.

Use a Certified **asthma & allergy friendly**® air cleaner.

To learn more, visit **aafa.org/iaq**

Asthma and Allergy Foundation of America

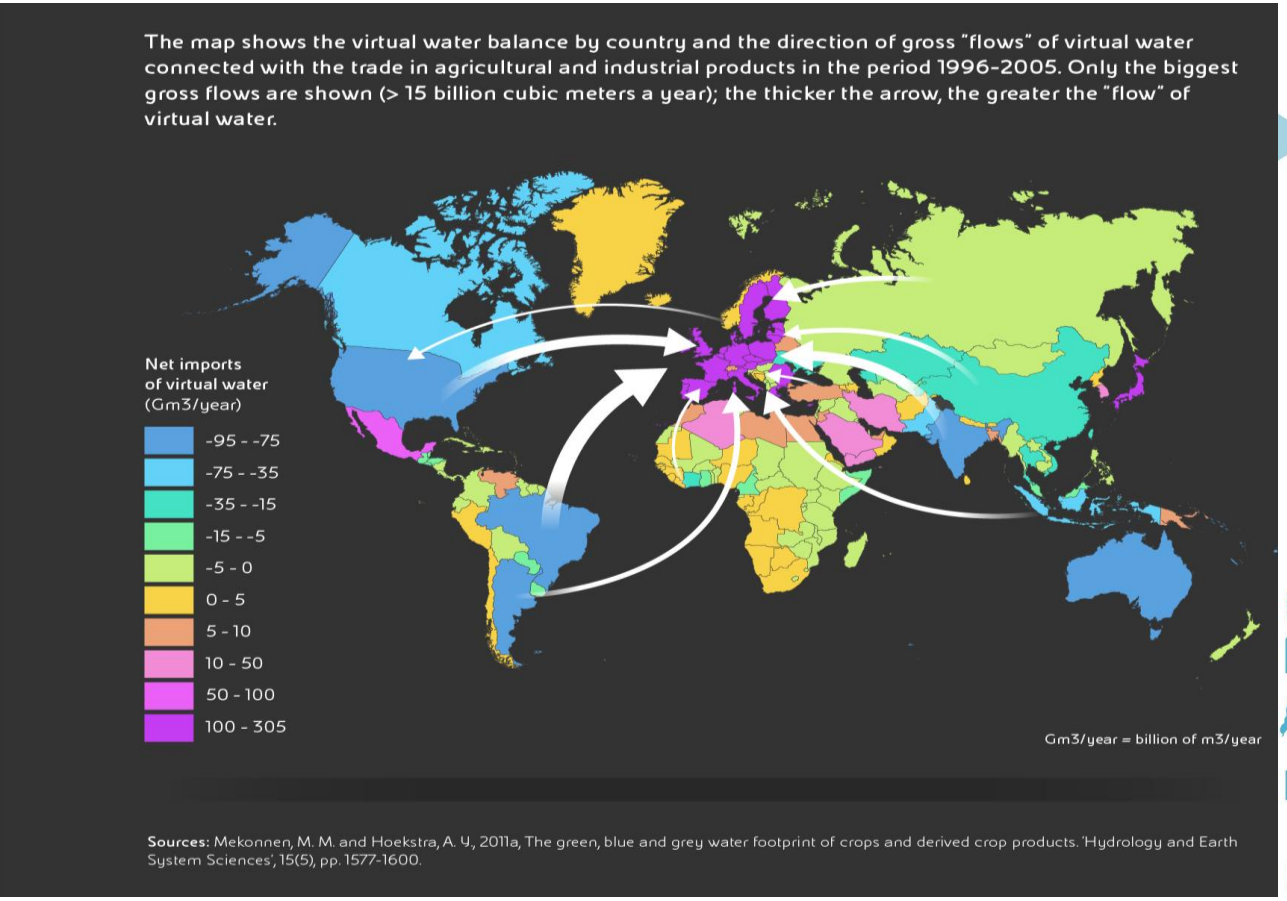
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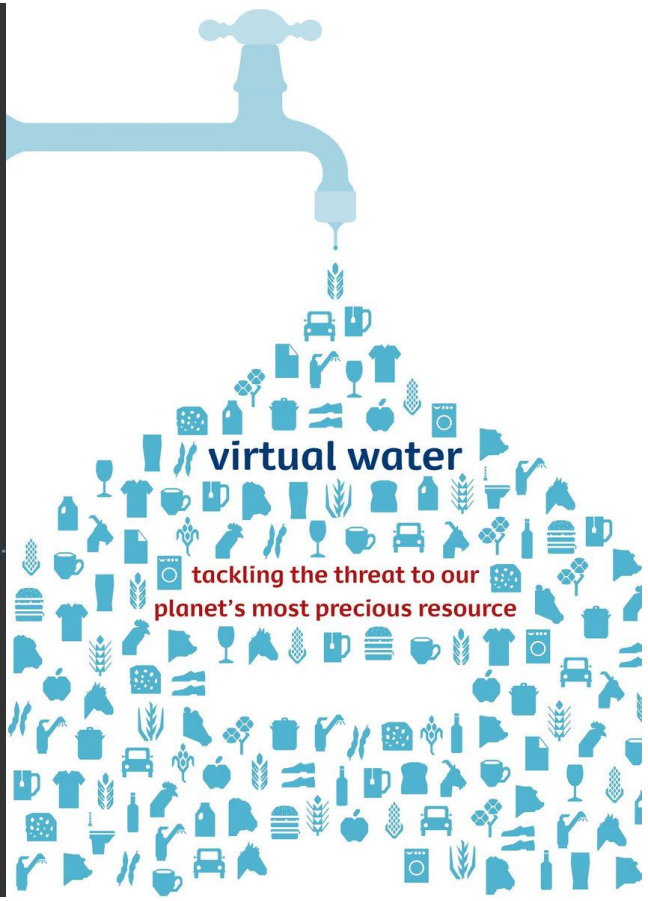
What makes Delhi's air so deadly??



The virtual water

Virtual water is the **water** used to create the goods and services that we consume and use.



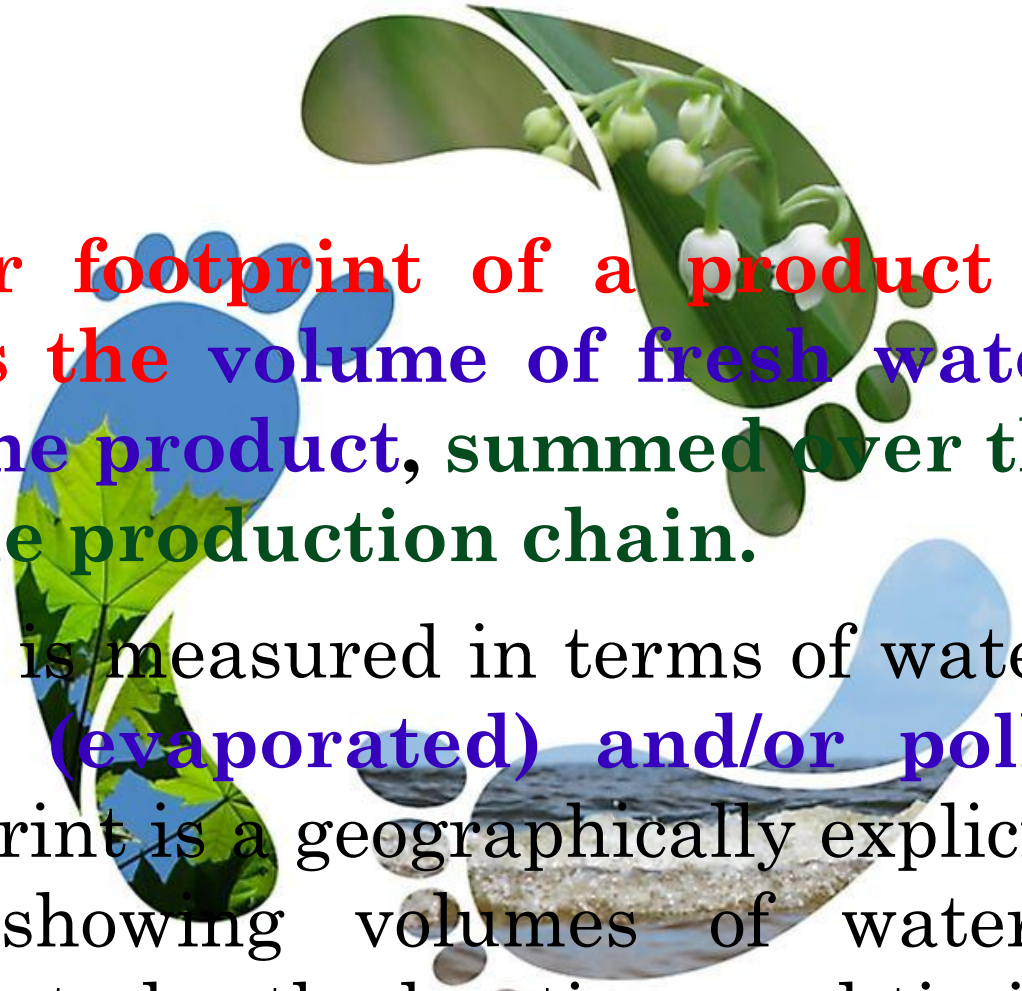


The virtual water

- While reading the paper in the morning, we are used to taking a cup of tea. To make a cup of tea, physically it require less than a cup of water. However, when we consider the amount of sugar, tea, and milk, which we add, although not physically visible, takes up 35 litres of water per cup, while one cup of coffee takes 140 liters of water to prepare. All this water, which is invisible in such items and goods is known as "Virtual Water". Every product we consume contains virtual water. In 1993, Professor John Allan (2008 Stockholm Water Prize Laureate), demonstrated the concept of virtual water which measures how water is embedded in the production and trade of food and consumer products.

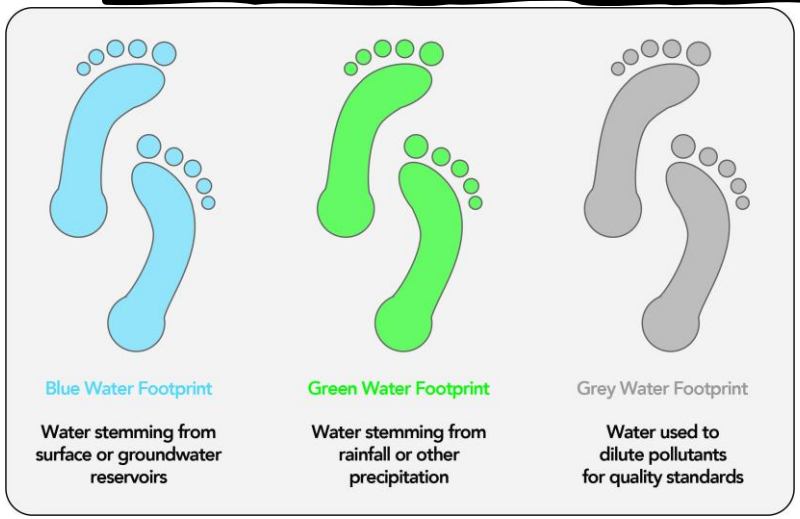
- **"Water footprint" is a useful tool to assess the amount of virtual water getting used/consumed in making different goods and services.**

Water footprint



- **The water footprint of a product (goods or service) is the volume of fresh water used to produce the product, summed over the various steps of the production chain.**
- 'Water use' is measured in terms of water **volumes consumed (evaporated) and/or polluted**. The water footprint is a geographically explicit indicator, not only showing volumes of water use and pollution, but also the locations and timing of water use.

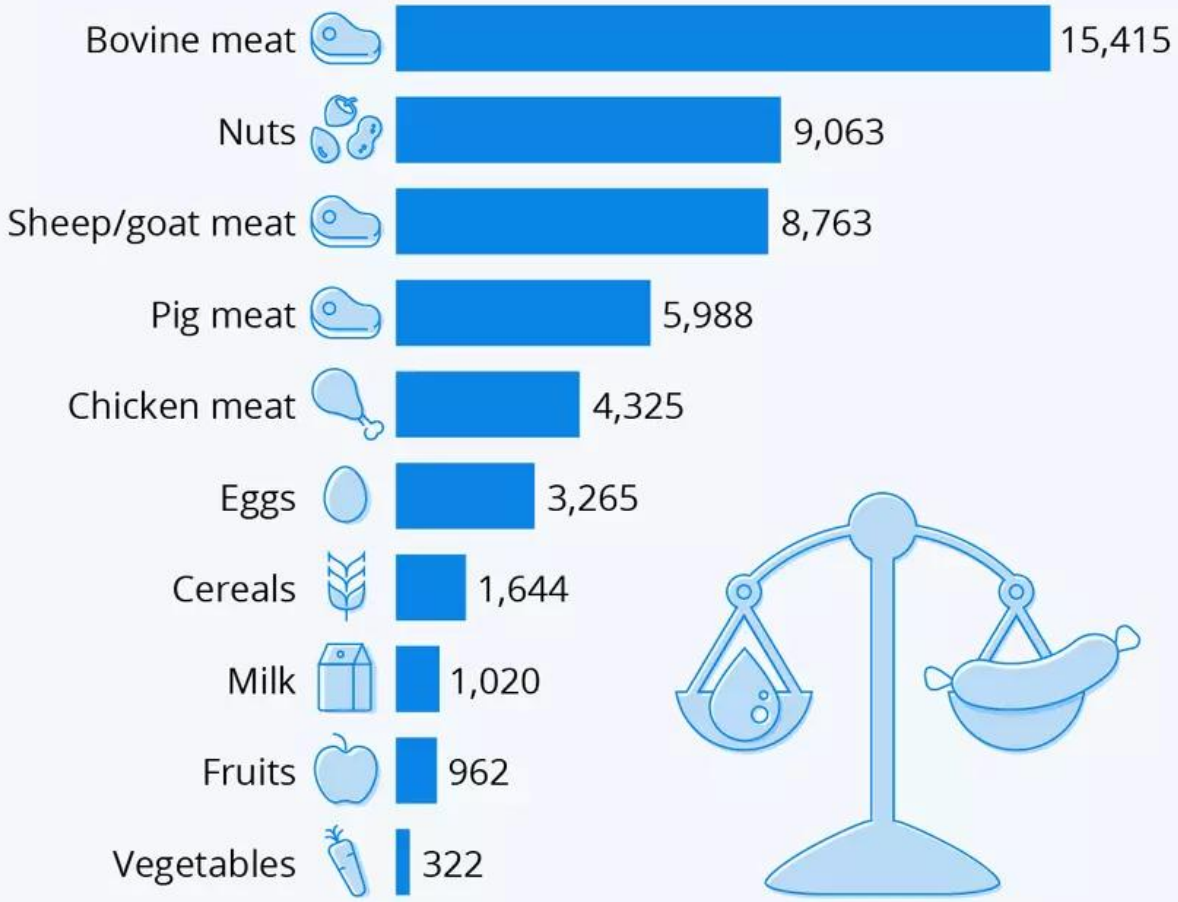
Water footprint



- **The blue water footprint** is the volume of freshwater that evaporated from the global blue water resources (**surface water and ground water**) to produce the goods and services consumed by the individual or community.
- **The green water footprint** is the volume of water evaporated from the global green water resources (**rainwater stored in the soil as soil moisture**).
- **The gray water footprint** is the **volume of polluted water** that associates with the production of all goods and services for the individual or community. This is calculated as the volume of water that is required to dilute pollutants to such an extent that the quality of the water remains above agreed water quality standards.

How Thirsty is Our Food?

Liters of water required to produce one kilogram of the following food products*





* Global averages

Source: Water Footprint Network

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statista


95% OF THE WATER WE USE EVERYDAY IS
INVISIBLE
COMBINED INVISIBLE WATER FOOTPRINT
28,429 LITRES
THIS IS MORE WATER THAN YOU HAVE DRUNK IN YOUR
ENTIRE LIFETIME
COTTON SUIT
14,763 LITRES
COTTON SHIRT
3,690 LITRES
MOBILE PHONE
912 LITRES
COFFEE, 200ML
140 LITRES
COTTON SOCKS
924 LITRES
LEATHER SHOES
8,000 LITRES
thirst

Water footprint

https://www.watercalculator.org/

Water Pollution

Water pollution is any change in water quality that can harm living organisms or make the water unfit for human uses such as drinking, irrigation, and recreation.

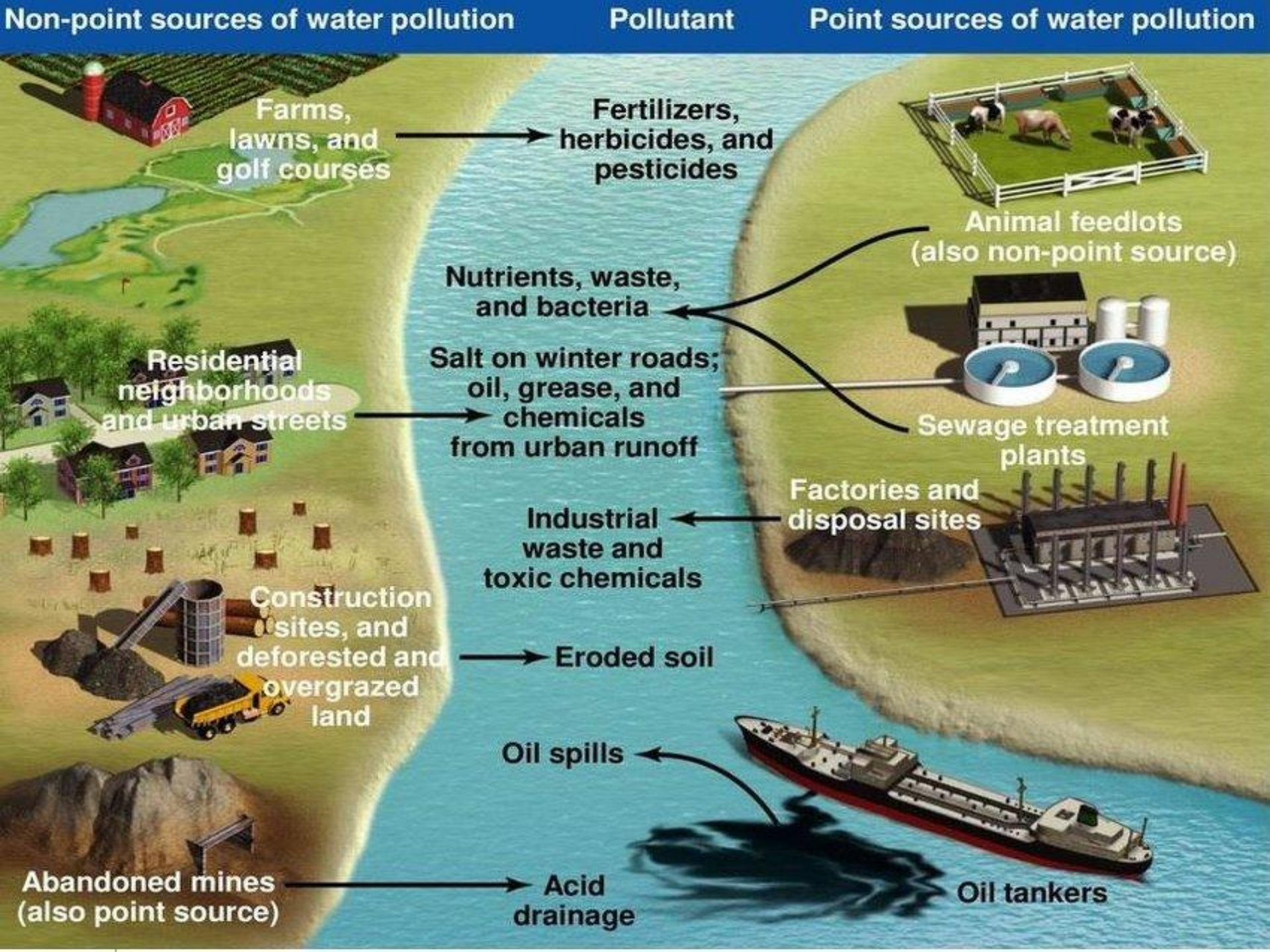
- It can come from single (point) sources or from larger and dispersed (non-point) sources.

Point sources discharge pollutants into bodies of surface water at specific locations through drain pipes, ditches, or sewer lines.
factories, underground mines, oil wells, and oil tankers.

Nonpoint sources are broad and diffuse areas where rainfall or snowmelt washes pollutants off the land into bodies of surface water.
runoff of eroded soil and chemicals such as fertilizers and pesticides from cropland, feedlots, logged forests, urban streets, parking lots, lawns, and golf courses.






Non-point sources of water pollution
Pollutant
Point sources of water pollution
Farms, lawns, and golf courses
Fertilizers, herbicides, and pesticides
Animal feedlots (also non-point source)
Nutrients, waste, and bacteria
Residential neighborhoods and urban streets
Salt on winter roads; oil, grease, and chemicals from urban runoff
Sewage treatment plants
Factories and disposal sites
Industrial waste and toxic chemicals
Construction sites, and deforested and overgrazed land
Eroded soil
Oil spills
Abandoned mines (also point source)
Acid drainage
Oil tankers

Eutrophication

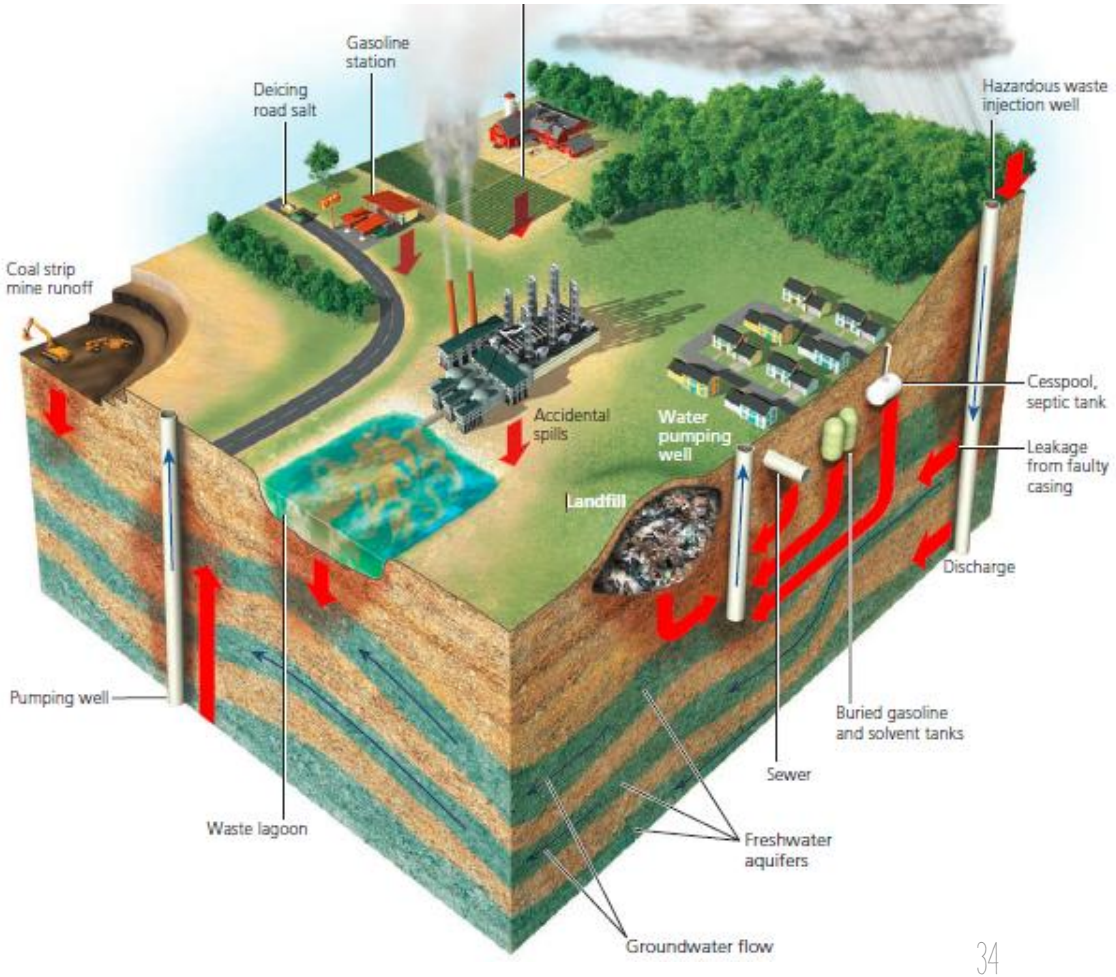




- **Eutrophication** is the name given to the natural nutrient enrichment of a shallow lake, a coastal area at the mouth of a river, or a slow-moving stream.
- It is caused mostly by runoff of plant nutrients such as nitrates and phosphates from land bordering such bodies of water.
- Over time, some lakes become more eutrophic as nutrients are added from natural and human sources in the surrounding watersheds.

Groundwater Pollution

Common pollutants such as fertilizers, pesticides, gasoline, and organic solvents can seep into groundwater from numerous sources



Is Bottled Water a Good Option?

Is your bottled water of higher quality than tap water?

Is your bottled water filtered?

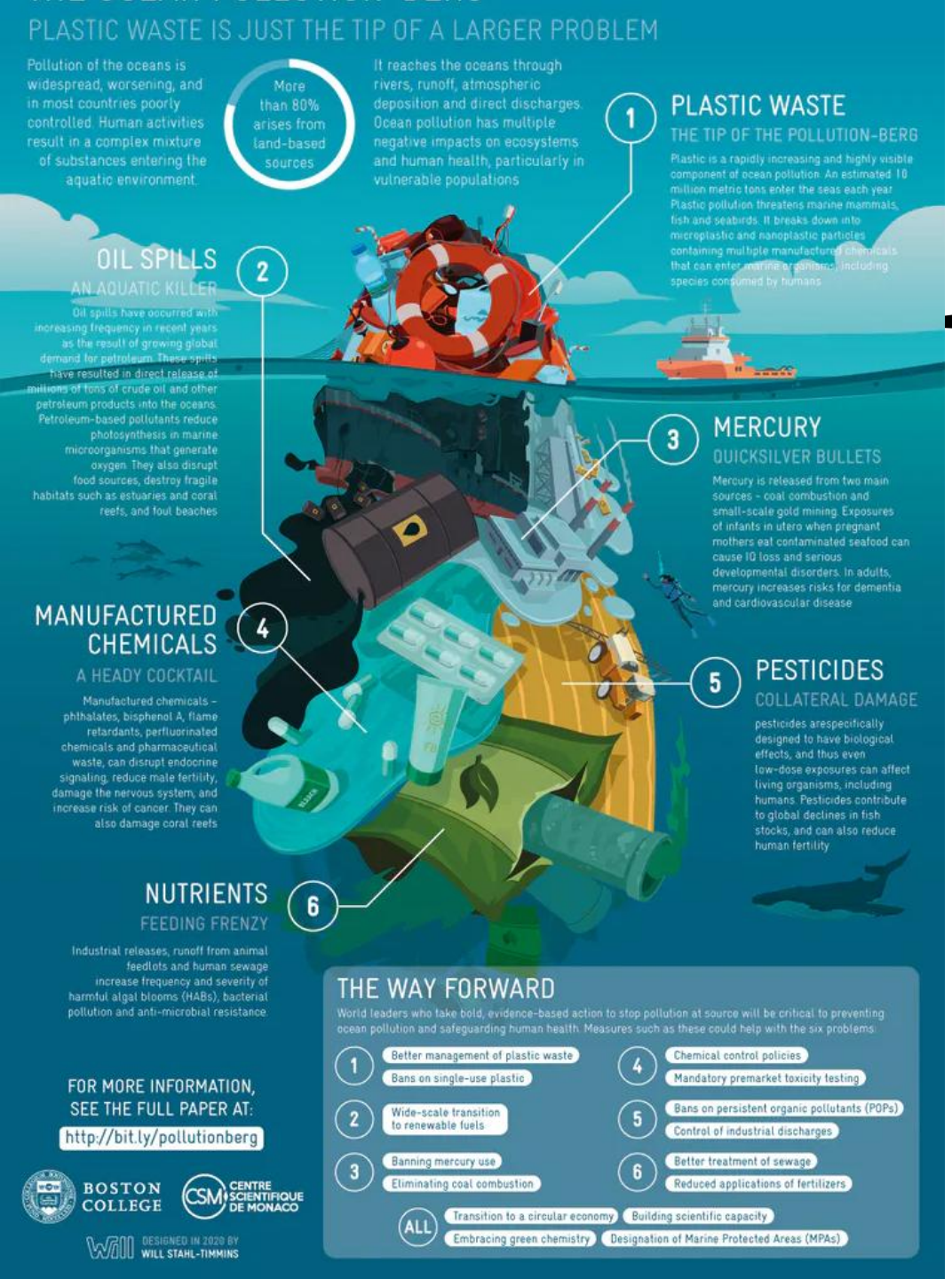


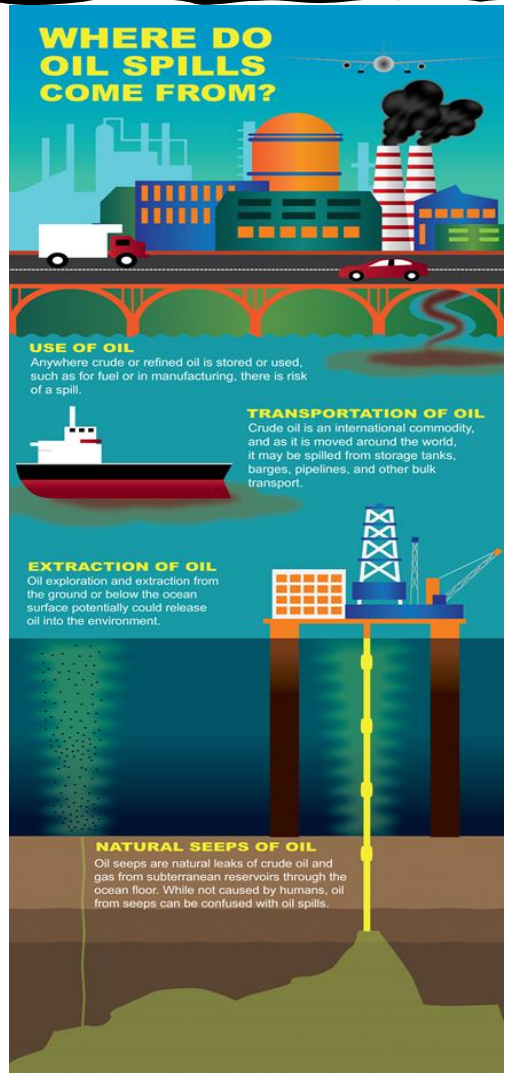
- Recent studies have found traces of phthalates, moulds, microbes, arsenic, and thousands of other contaminants in bottled water.

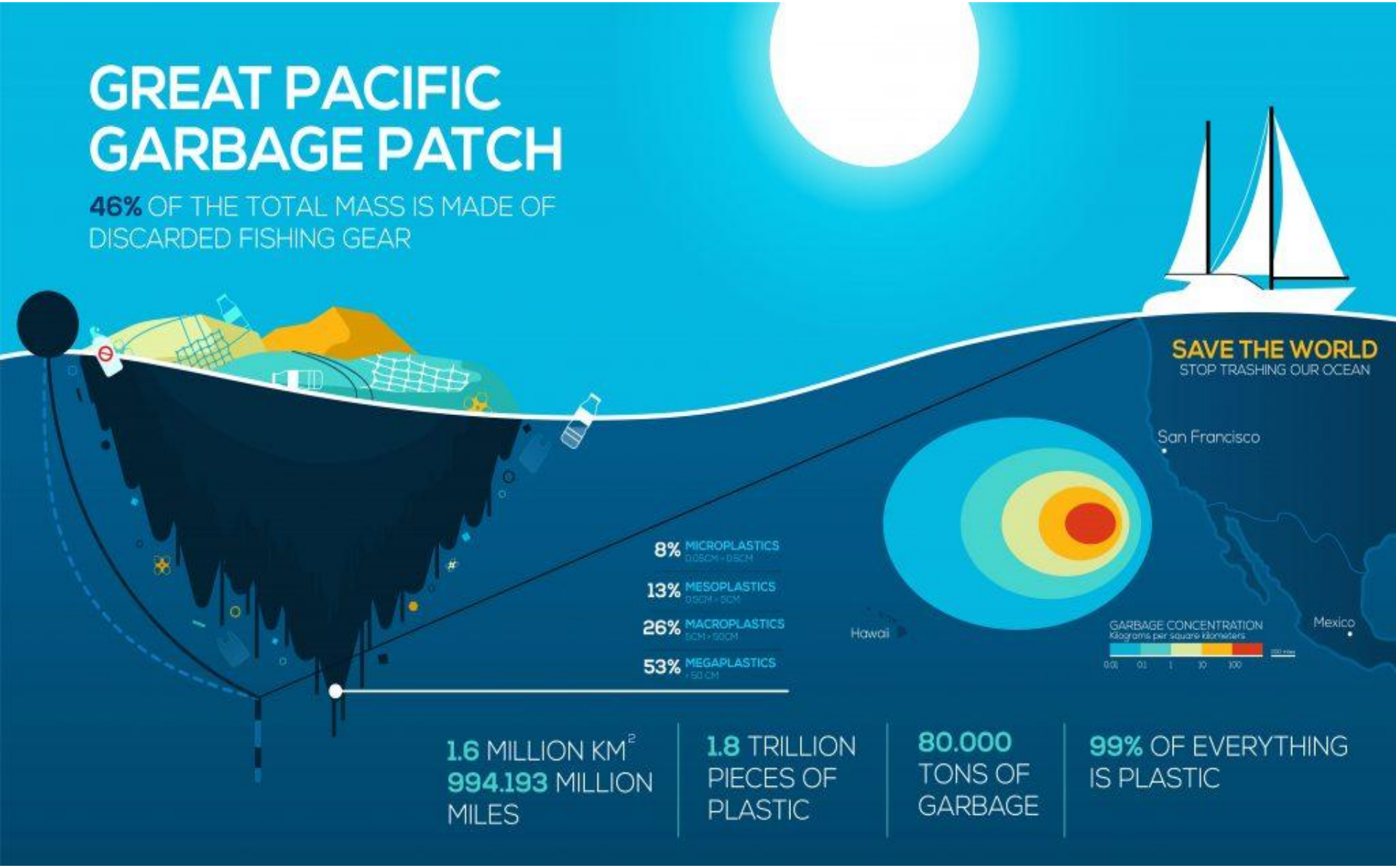
- BPA and other plastic toxins can then make their way into your bloodstream, which can cause a host of problems including various cancers as well as liver and kidney damage.
- It has been found that people who drink a lot of bottled water, and in particular those who had high concentrations of BPA in their urine, were three times more likely to suffer from cardiovascular disease, and were 2.4 times more likely to develop Type 2 diabetes than individuals with low concentrations of BPA in their bodies.
- World Health Organisation (WHO) study found that in 93% of popular bottled water brands tested the water contained plastic fibers.
- Can cause increase in body weight.
- Reusing can also pose a problem.

Nowadays, Plastic bottles do not contain BPA or BPS but PET (polyethylene terephthalate)

Ocean Pollution






GREAT PACIFIC GARBAGE PATCH
46% OF THE TOTAL MASS IS MADE OF DISCARDED FISHING GEAR
SAVE THE WORLD
STOP TRASHING OUR OCEAN
San Francisco
Hawaii
Mexico
GARBAGE CONCENTRATION
Kilograms per square kilometers
8% MICROPLASTICS
13% MESOPLASTICS
26% MACROPLASTICS
53% MEGAPLASTICS
1.6 MILLION KM²
994.193 MILLION MILES
1.8 TRILLION PIECES OF PLASTIC
80.000 TONS OF GARBAGE
99% OF EVERYTHING IS PLASTIC

Microplastics = Macro-problem!!



Control water pollution

Coastal Water Pollution

Prevention

Separate sewage and storm water lines

Require secondary treatment of coastal sewage

Use wetlands and other natural methods to treat sewage

Ban dumping of wastes and sewage by ships in coastal waters

Strictly regulate coastal development, oil drilling, and oil shipping

Require double hulls for oil tankers

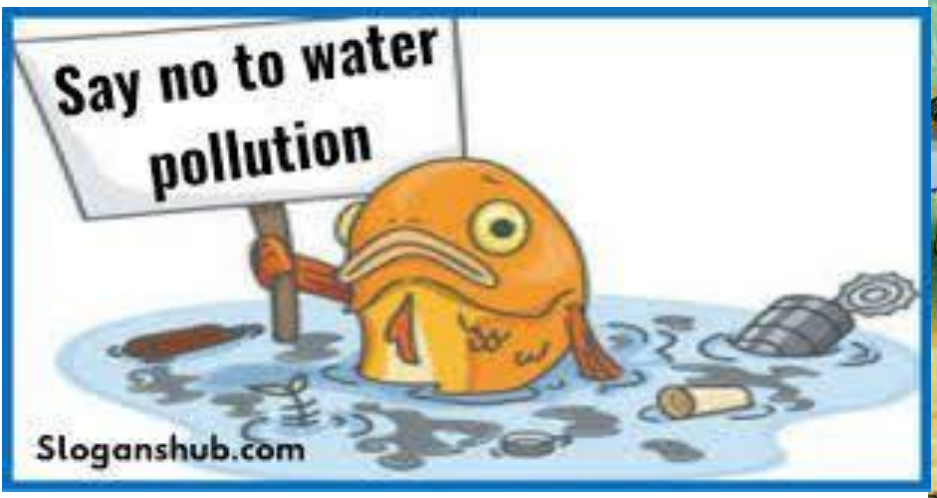




Cleanup

Improve oil-spill cleanup capabilities

Use nanoparticles on sewage and oil spills to dissolve the oil or sewage (still under development)





- Prevent groundwater contamination
- Reduce nonpoint runoff
- Work with nature to treat sewage and reuse treated wastewater
- Find substitutes for toxic pollutants
- Practice the four Rs of resource use (refuse, reduce, reuse, recycle)
- Reduce air pollution
- Reduce poverty
- Slow population growth

- Fertilize garden and yard plants with manure or compost instead of commercial inorganic fertilizer
- Minimize use of pesticides, especially near bodies of water
- Prevent yard wastes from entering storm drains
- Do not use water fresheners in toilets
- Do not flush unwanted medicines down the toilet
- Do not pour pesticides, paints, solvents, oil, antifreeze, or other harmful chemicals down the drain or onto the ground





Why It's Important To Conserve Water



To prepare for future droughts

To guard against rising costs & potential conflict





To preserve the environment





To make water available for recreational purposes

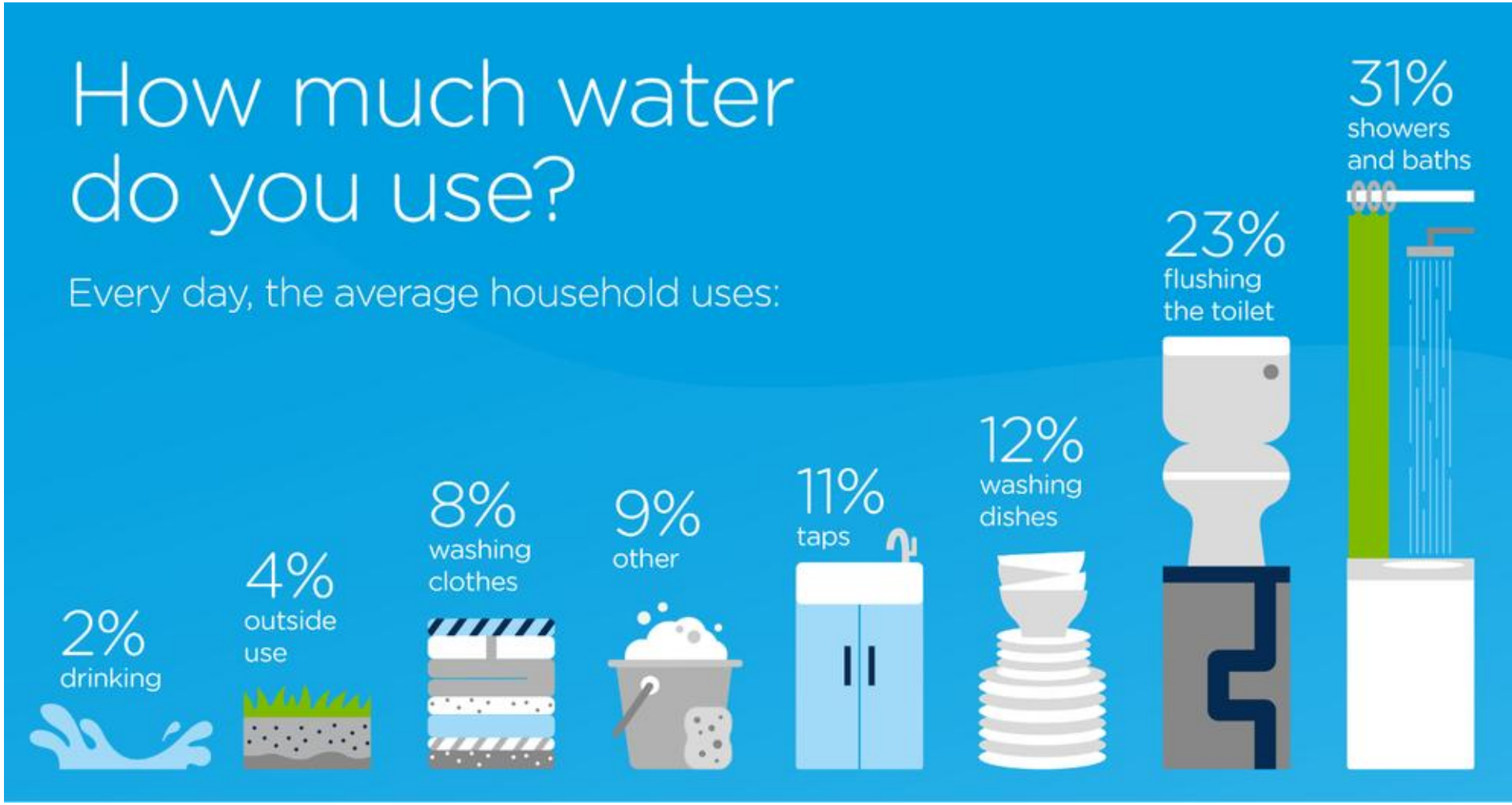


To strengthen communities



the balance

Water Use


How much water do you use?
Every day, the average household uses:
2% drinking
4% outside use
8% washing clothes
9% other
11% taps
12% washing dishes
23% flushing the toilet
31% showers and baths

Ways to Conserve Water

OUTDOORS

1 WATER EARLY OR LATE
Before 8am or after 7pm reduces evaporation

2 WATER DEEPLY BUT INFREQUENTLY
One or two deep waterings is better for plants than several shallow waterings

and

GET WATER TO THE ROOTS
Hand watering, drip irrigation, or soaker hoses do this well

3 WASH VEHICLES
at locations that **recycle** the water

4 USE A BROOM
instead of a hose to clean sidewalks, driveways, and patios

INDOORS

6 TEST FOR TOILET LEAKS
at least once a year—these can be silent!

To check for toilet leaks: Drop food coloring into the tank, leave overnight. If the dye seeps into the bowl, you have a leak.

7 WASH FULL LOADS
in clothes and dishwasher

Replacing appliances? Consider **water-efficient models for long-term savings.**
NUD helps sponsor rebates on water-efficient models: see savingwater.org/rebates

8 TURN OFF THE WATER
while you brush your teeth or shave—save 4 gallons a minute, **200 gallons a week** for a family of four.

9 REDUCE SHOWER TIME
By 1-2 min

Use NUD's **free 5-min shower timers!**

LEAKS

10 LOOK OUT FOR LEAKS
inside & outside

One drip every second adds up to **five gallons per day of water lost!**

Obvious leaks:
- Faucets
- Hose bibs

Sneaky leaks:
- Toilet leaks
- Underground irrigation leaks

Learn how to detect & troubleshoot leaks at:
www.nud.net/how-do-i/find-and-fix leaks

Find video tutorials on fixing common leaks at:
Savingwater.org/indoors/fixingleaks

BUSINESS

Specific water-saving tips for businesses at savingwater.org

Practise rain water harvesting at home		Keep water stored in buckets and tubs in the wash room for use
Attend to repairs in the bathroom as and when required		Attend to leaky faucets
Do all the laundry at one go; avoid frequent washes		Go in for short showers, during summer
Talk to friends and neighbours about saving water		Organise awareness campaigns in your neighbourhood on ways to save water
Close all taps and valves before you step out of the house or go on a vacation		Fill half a glass of water to drink first; refill, if required

Water Conservation: Engage, Conserve, and Restore



- Decreasing run-off losses :

  Contour cultivation – furrows and ridges

- Reducing evaporation losses

A copolymer of Starch and acrylonitrile –"super slurper" can absorb water





- Storing water in soil

- Reducing irrigation losses

  Sprinkling irrigation





- Re-use of water

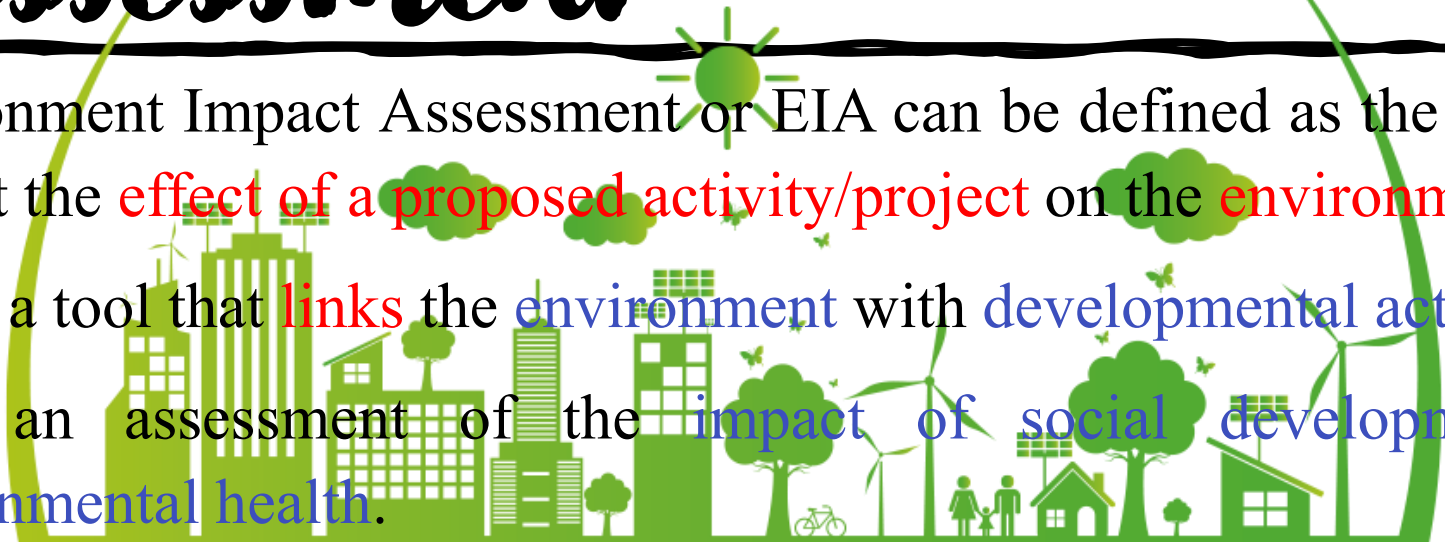
- Preventing wastage of water

Bhopal Gas Tragedy:

"500 dollars is plenty good for an Indian!"

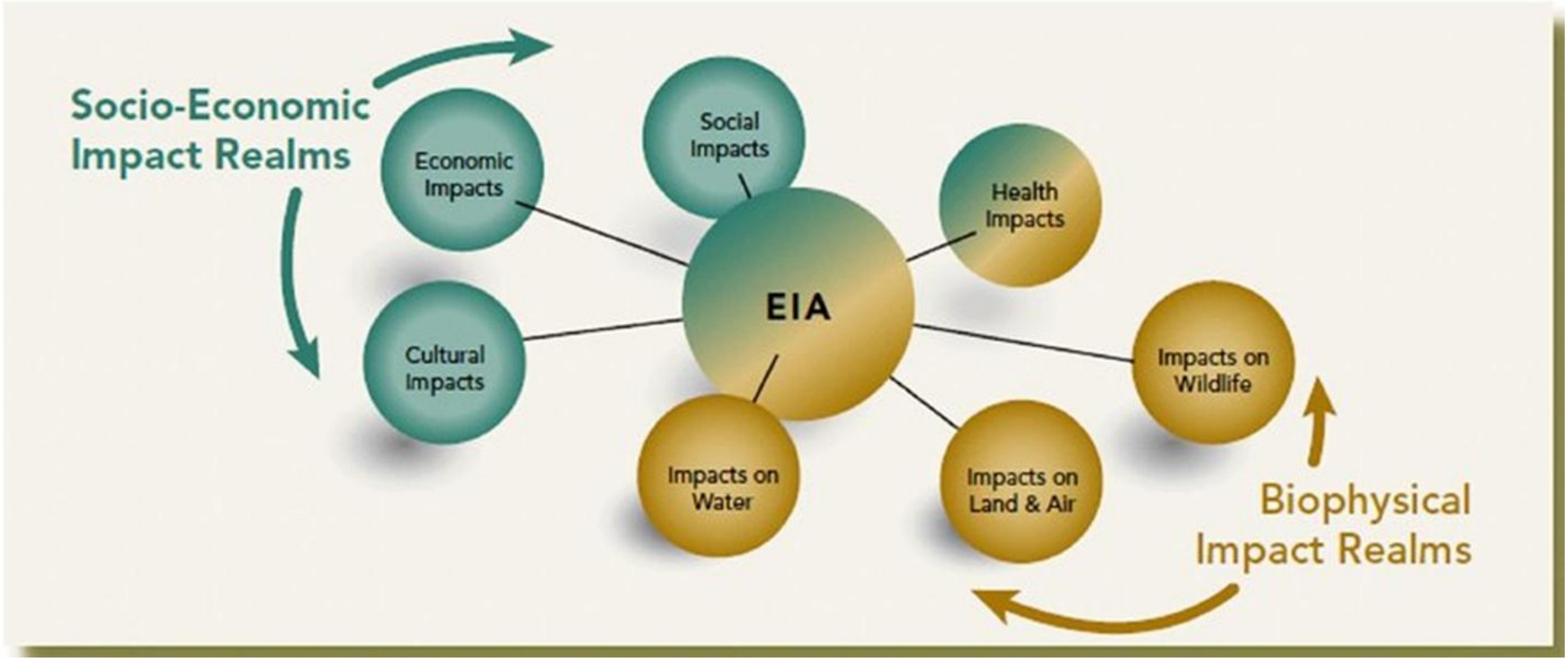


Environment Impact Assessment



- Environment Impact Assessment or EIA can be defined as the study to predict the effect of a proposed activity/project on the environment.
- EIA is a tool that links the environment with developmental activities.
- It is an assessment of the impact of social development on environmental health.
- It ensures that the development of a project is environmentally sound and sustainable.
- The Ministry of Environment and Forest under the Environmental Protection Act of 1986, initiated the **Environmental Impact Assessment (EIA) in 1994**. It was reengineered in 2006.

Important aspects of EIA



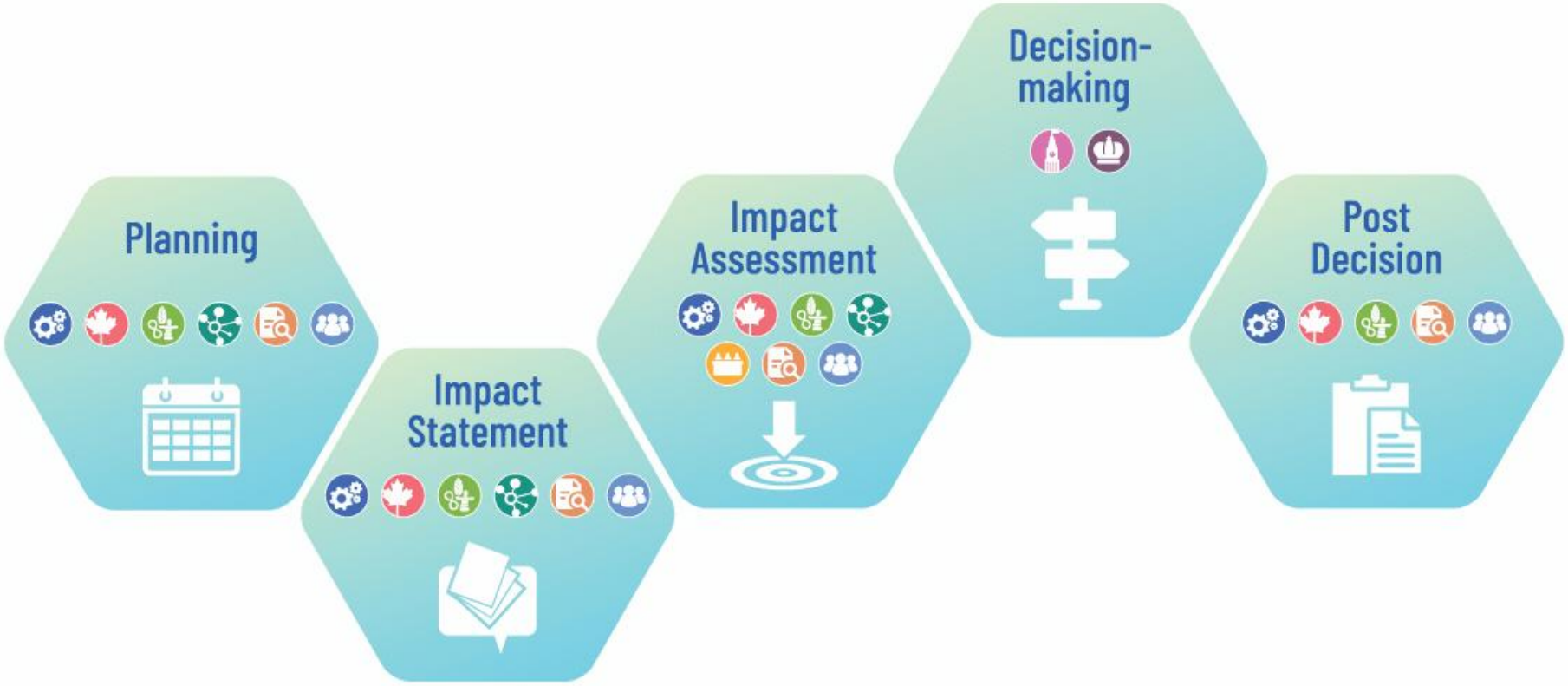
IMMEDIATE OBJECTIVES

- Improve the environmental design of the proposal.
- Check the environmental acceptability of the proposals compared to the capacity of the site and the receiving environment.
- Ensure that the resources are used appropriately and efficiently.
- Identify appropriate measures for mitigating the potential impacts of the proposal.
- Facilitate informed decision making, including setting the environmental terms for implementing the proposal.

LONG TERM OBJECTIVES

- Avoid irreversible changes and serious damage to the environment .
- Safeguard valuable resources, natural areas and ecosystem components.
- Enhance the social aspects of proposals.
- Protect human health and safety.
- To provide awareness about environmental values/ethics.

Environmental Impact Assessment



ENVIRONMENTAL CLEARANCE

- **Environmental clearance** granted by the EIA Agency in the Ministry of Environment and Forests (MoEF), Government of India.

- The **Environmental Clearance** process is required for 39 types of projects and covers aspects like screening, scoping and evaluation of the upcoming project.

- The main purpose is to assess the impact of the planned project on the environment and people and to try to abate/minimise the same.

ENVIRONMENTAL CLEARANCE CATEGORIES:

- **<u>Category A projects</u> :** At Central level MOEF, based on decisions of Expert Appraisal Committee (EAC)- National Level

- **<u>Category B projects</u> :** At state level, the State Environment Impact Assessment Authority (SEIAA), based on decision of a State level Expert Appraisal Committee (SEAC) as to be constituted for in this notification – State Level

ENVIRONMENTAL CLEARANCE CATEGORIES:

Industry	Category A	Category B
Mining of minerals	≥ 50 ha. of mining lease area	< 50 ha. And ≥ 1 ha. of mining lease area
Thermal power plants	≥ 500 MW	< 500 MW
Cement	≥ 1 M tons per year	< 1 M tons per year
Metallurgical industries	Primary metallurgical processing	Secondary metallurgical processing

Self-study Portion:

ENVIRONMENT PROTECTION ACT, 1986

- The act covers all forms of pollution ; air, water , soil and noise.
- It provides the safe standards for the presence of various pollutants in the environment.
- It prohibits the use of hazardous material unless prior permission is taken from the central government.
- It allows the central government to assign authorities in various jurisdictions to carry out the laws of this acrt

THE AIR ACT

The Air (Prevention and Control of Pollution) Act of 1981, or the Air Act, in short, was a law passed by the Parliament of India to prevent and control the harmful effects of air pollution in India. This act is seen as the first concrete step taken by the government of India to combat air pollution.

The key features of the Act include:

- Advising Central Government of Air and Air Pollution related issues
- Research about the causes and impact of Air Pollution
- Spread awareness to stop air pollution
- To establish central and State Boards and empower them to monitor air quality and control pollution

THE WATER ACT

- Water act , 1974 is a comprehensive legislation that regulates agencies responsible for checking on water pollution and ambit of pollution control boards both at the centre and states.
- The water act, 1974 was adopted by the parliament with the aim of prevention and control of water pollution in india.
- One of the important provision of the water act, 1974 is to maintain and restore the wholesomeness of our aquatic resources.
- Under water act, 1974 , pollution control boards were created, who are responsible for implementation of its provisions.

WILD LIFE PROTECTION ACT 1972

- Prohibition of Hunting : Any wild animal killed or wounded in defense of any person shall be Government property.
- Declaration of Sanctuary : The State Government declare any area as Sanctuary for the purpose of protecting, propagating or developing wildlife or its environment.
- Declaration of National Parks : an area, by reason of its ecological, faunal, floral, geomorphological for the purpose of protecting, propagating or developing wildlife therein or its environment.
- Declaration of closed area : State Government, by notification, declare any area closed to hunting for such period as may be specified in the notification.

FOREST CONSERVATION ACT, 1980

- The Forest Conservation Act 1980 was enacted to help conserve the country's forests.
- It strictly restricts and regulates the de-reservation of forests or use of forest land for non-forest purposes without the prior approval of Central Government.
- To this end the Act lays down the prerequisites for the diversion of forest land for non-forest purposes.



16TH FEBRUARY – THE KYOTO PROTOCOL (2005)



192 COUNTRIES

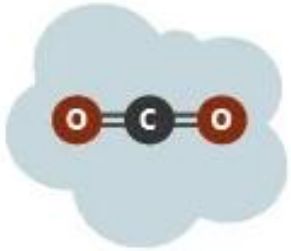
- signed and ratified
- ratified but withdrawn
- signed but not ratified

TARGET
↓5.2%
by 2012 relative to 1990

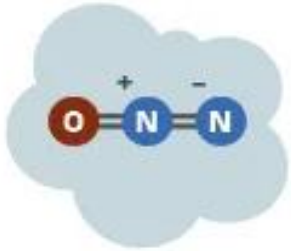




METHANE



CARBON DIOXIDE



NITROUS OXIDE

GREENHOUSE GASES

- HYDROFLUOROCARBONS
- PERFLUOROCARBONS
- SULFUR HEXAFLUORIDE

The Kyoto Protocol is an international treaty to reduce the emission of six greenhouse gases. 192 countries are party to the protocol, but only 38 had binding targets for the first period (2008–2012). These 38 countries met their target collectively – but an absence of binding targets for developing nations such as China and India meant that worldwide greenhouse gas emissions continued to rise.

Greenhouse gas emissions



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The **Montreal Protocol on Substances that Deplete the Ozone Layer:** a protocol to the Vienna Convention for the protection of the ozone layer is an international treaty designed to protect the ozone layer by phasing out the production of numerous substances that are responsible for ozone depletion. It was agreed on **16 September 1987**, and entered into force on 1 January 1989.

HFCs ARE POWERFUL GREENHOUSE GASES

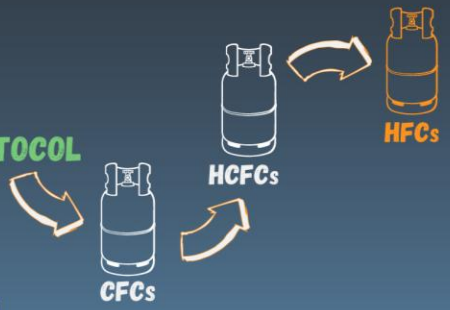
THEY ARE USED IN



REFRIGERATION & AIR-CONDITIONING

USE IS GROWING RAPIDLY

AS THEY REPLACE OZONE-DEPLETING SUBSTANCES BEING PHASED OUT BY THE MONTREAL PROTOCOL



1.6 BILLION

AIR-CONDITIONING UNITS WILL BE INSTALLED WORLDWIDE BY 2050, MAKING THE PROBLEM EVEN WORSE



LUCKILY, SUSTAINABLE AND ENERGY-EFFICIENT ALTERNATIVES ARE AVAILABLE, INCLUDING NATURAL REFRIGERANTS

MONTREAL PROTOCOL

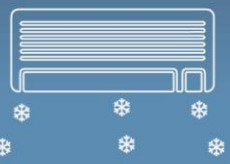
IS THE WORLD'S MOST SUCCESSFUL ENVIRONMENTAL AGREEMENT

AN AMBITIOUS AGREEMENT TO REDUCE HFC USE VIA THE MONTREAL PROTOCOL CAN AVOID

100 BILLION

TONNES OF CO_2e BY 2050 & UP TO 0.5°C WARMING BY 2100



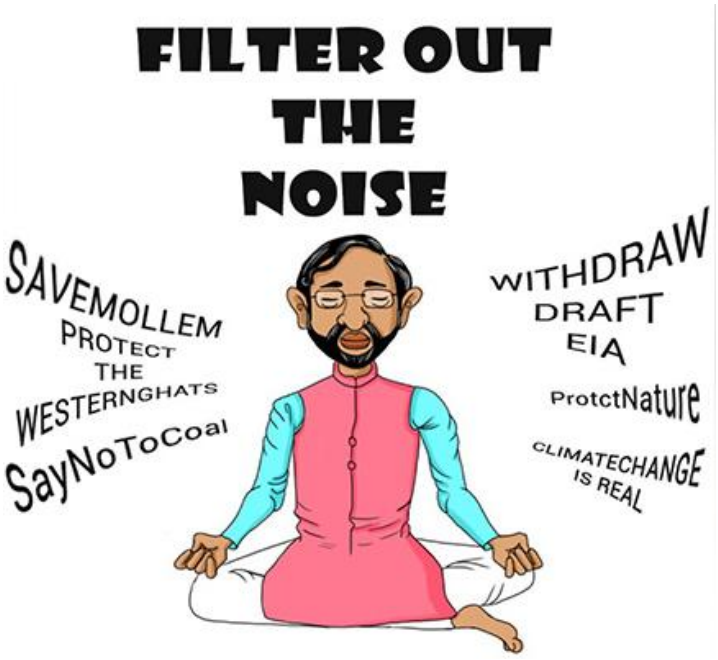


COOLING WITH HFCs

NOT SO COOL ANYMORE

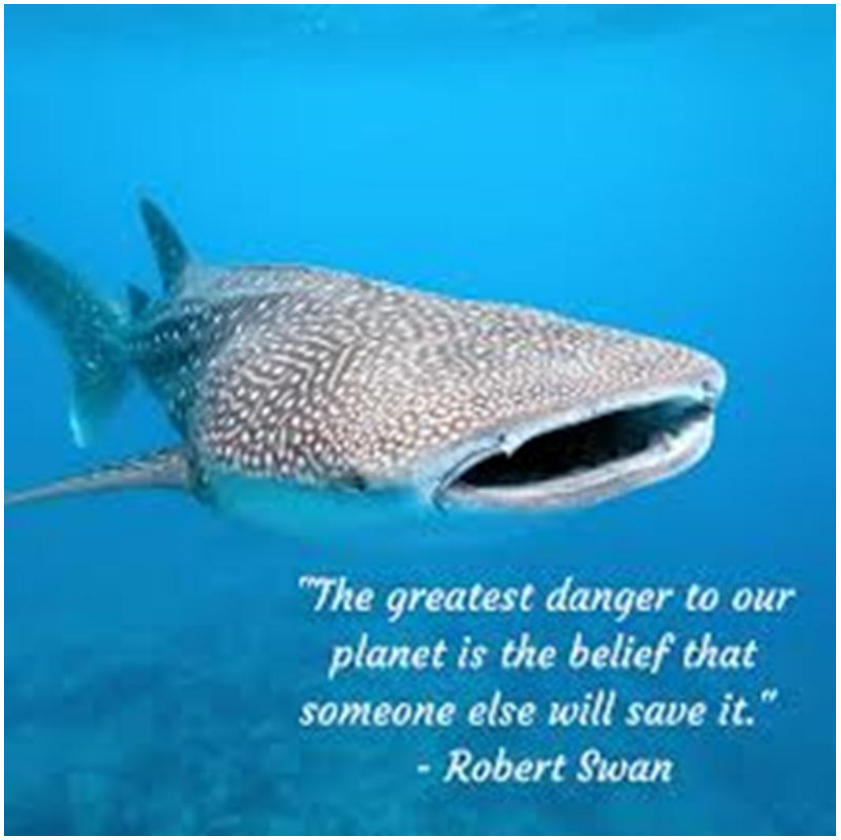

Refuse
I will refuse disposable and single use plastics like plastic straws and plastic bags as much as I can.
Reduce
I will reduce my plastic waste by making greener purchasing decisions and consider the lifecycle of the plastics which I bring into my life.
Recycle
I will recycle and dispose the plastics responsibly. Use recycle bins and encourage those I know to do so too.
Re-use
I will reuse old plastics instead of buying new ones.
Remind
I will remind others about the need to follow the 4Rs, and reconsider their actions.


Sustainable Development
PEOPLE
End poverty and hunger in all forms and ensure dignity and equality
PROSPERITY
Ensure prosperous and fulfilling lives in harmony with nature
PEACE
Foster peaceful, just and inclusive societies
PARTNERSHIP
Implement the agenda through a solid global partnership
PLANET
Protect our planet's natural resources and climate for future generations


FILTER OUT THE NOISE
SAVEMOLLEM
PROTECT THE WESTERNGHATS
SayNoToCoal
WITHDRAW DRAFT EIA
ProtctNature
CLIMATECHANGE IS REAL


NUMBER OF POLITICIANS
WHO THINK ENVIRONMENT IS A SERIOUS ISSUE


UP NEXT FOR AUCTION: FORESTS IN THE HEART OF INDIA!
BID GENEROUSLY, MINER FRIENDS! AND LEAVE ENVIRONMENTAL CLEARANCE TO THE MoEFCC.


"The greatest danger to our planet is the belief that someone else will save it."
- Robert Swan


Thank you!